Amplitude-Robust Qubit Spectroscopy Using Echo-Root-Lorentzian Pulse Shapes

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We present a spectroscopy method for driven two-level systems based on a root-Lorentzian envelope with a π-phase inversion at its peak. For driven-spectroscopy pulses shorter than the lifetime T_1, the resulting echo-like symmetry cancels on-resonant Rabi oscillations while preserving the off-resonant spectral response and suppressing conventional power broadening. Measurements on a superconducting circuit show a narrow echo feature over more than two decades of peak drive, extending the amplitude range over which near-T_2-limited spectroscopy can be performed. Across peak Rabi frequencies up to 50 MHz, the measured response remains centered near resonance without the monotonic AC-Stark displacement expected for the constant-drive reference. Three-level simulations show that combining a smoothed inversion and derivative removal by adiabatic gate (DRAG) with polynomial edge correction improves high-drive linewidth and probability contrast, at the cost of a broader low-drive line. A direct comparison of measured spectra with numerical calculations validates the model and supports a practical approach to amplitude-robust, high-resolution spectroscopy.

Determining the transition frequency of a quantum system is a fundamental task in spectroscopy, quantum metrology, and quantum sensing. The simplest setting is a two-level system, whose energy splitting defines a spectroscopic resonance and can encode changes in its local environment. This frequency is commonly estimated by sweeping a coherent drive across the transition and monitoring the resulting population response [1–4]. Such measurements underlie both the characterization of quantum devices and the use of quantum systems as probes of external perturbations.

In direct population spectroscopy, spectral resolution and measurement robustness are competing requirements. Conventional continuous-wave spectroscopy generally applies a drive long enough for the population to approach a steady state. In the weak-drive regime, decoherence produces a homogeneous linewidth that scales as $1/T_2$, establishing the natural coherence-limited resolution scale of the transition. Increasing the drive strength makes the population response more pronounced, but also power broadens the transition. Conventional high-resolution spectroscopy therefore requires the drive amplitude to be chosen carefully, balancing a pronounced population response against power broadening. High-amplitude spectroscopy is further complicated when the nominal two-level system is embedded in a multilevel spectrum. Off-resonant coupling to neighboring transitions can produce amplitude-dependent AC-Stark shifts, line-shape distortions, and leakage, causing the apparent resonance to move with drive strength [5–7].

Ramsey interferometry provides a standard alternative to direct spectroscopy and, in principle, enables frequency estimation at the T_2 limit [2]. Other sensing protocols can surpass this probe-T_2 scale using inter-shot correlations, external references, dynamical decoupling, or tailored filter functions [3, 4, 8–11]. Ramsey generally presupposes an approximate transition frequency and calibrated $\pi/2$ pulses for state preparation and readout, and is therefore not fully calibration independent. Achieving T_2-limited resolution also requires long interrogation times and sufficiently dense temporal sampling to resolve the resulting interference fringes. Finite-pulse imperfections and probe-induced frequency shifts, including unintended AC-Stark shifts during the control pulses, can nevertheless bias the extracted transition frequency [12, 13].

Here, we propose a spectroscopic method for probing the qubit transition frequency using an echo-root-Lorentzian drive pulse. Previous theoretical work showed that Lorentzian pulses exhibit power narrowing, complementary to the conventional power broadening observed for square (constant-amplitude) pulses, as a consequence of their adiabatic properties [14]. Power narrowing with powers-of-Lorentzian pulses was subsequently demonstrated experimentally on IBM Quantum, including the cutoff broadening introduced by truncating the pulse wings [15]. In this work, we extend this framework to longer pulse durations approaching the T_1 timescale and introduce a π-phase jump at the pulse peak. The resulting echo property effectively cancels the net pulse area while preserving the adiabatic properties of the drive. This phase inversion reverses the coherent phase accumulation at resonance, suppressing on-resonant excitation while maintaining an off-resonant spectral response across the frequency domain, and thus making resonance distinguishable.

We define the echo-root-Lorentzian drive from the order-n generalized Lorentzian envelope as

$$L^{(n)}(t) = \frac{1}{\left[1 + \left(\frac{t}{\tau}\right)^2\right]^n}, \tag{1}$$

$$L_{\text{echo}}(t) := -L^{(1/2)}(t)\,\text{sgn}(t). \tag{2}$$

where τ sets the characteristic pulse duration and n controls the algebraic decay of the wings.

For the untruncated, plain generalized Lorentzian with

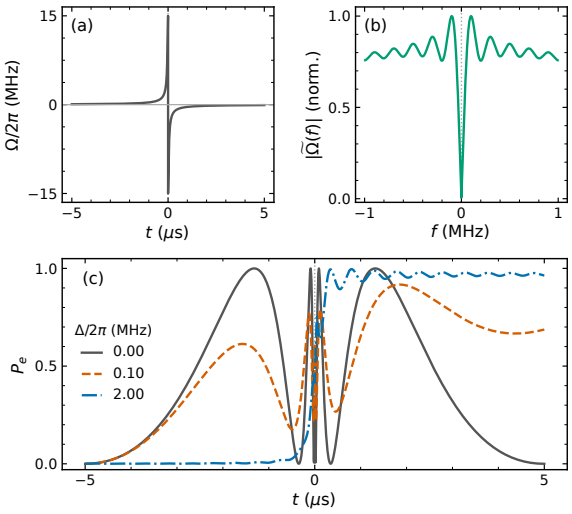


Figure 1. Pulse waveform and echo-cancellation mechanism. (a) Signed echo-root-Lorentzian drive envelope with peak amplitude $\Omega_0/(2\pi) = 15$ MHz and a π phase inversion at the pulse midpoint. (b) Normalized magnitude of the waveform Fourier spectrum, showing the spectral null at zero frequency produced by the phase inversion. (c) Unitary two-level excited-state population during the pulse at $\Delta/(2\pi) = 0$, 0.10, and 2.00 MHz, without relaxation or dephasing. At resonance, the evolution accumulated during the first half is exactly reversed during the second half, whereas finite detuning prevents complete cancellation.

$n > 1/2$, the boundary of adiabatic following provides an estimate of the characteristic spectral width. Applying the usual adiabatic condition to this envelope gives [14]

$$\Delta_b \tau \propto (\Omega_0 \tau)^{-\frac{1}{2n-1}}, \tag{3}$$

where Δ_b is the angular detuning at which adiabatic following ceases and Ω_0 is the peak angular Rabi frequency. Because the exponent is negative, this characteristic detuning decreases as the drive strength increases, producing power narrowing rather than conventional power broadening. The exponent grows without bound as $n \to 1/2^+$, so within its domain this asymptotic result predicts the steepest narrowing trend at the root-Lorentzian boundary. At $n = 1/2$, finite truncation, decoherence, and the echo phase inversion regularize the response, and the FWHM Γ_f used below is therefore obtained from the complete experimental spectrum. This interpretation is consistent with the power narrowing and cutoff broadening observed for powers of Lorentzian pulses by Mihov and Vitanov [15]. In contrast, a square pulse exhibits conventional power broadening, with $\Gamma_\omega \propto \Omega_0$ at large drive. The derivation and the marginal $n = 1/2$ limit are detailed in the Supplemental Material [16].

The constant-drive reference in Fig. 2 illustrates linear power broadening at large drive. In contrast, the root-Lorentzian pulse displays the characteristic power-narrowing behavior and maintains a narrow linewidth. However, even for pulse durations shorter than the qubit lifetime T_1, coherent oscillations arise for root-Lorentzian pulses, resulting in amplitude-dependent linewidth variations that hinder precise spectral extraction.

Introducing a π phase inversion at the peak of the root-Lorentzian envelope partitions the drive into two sequential segments of opposite phase. For an ideal symmetric unitary two-level pulse, the corresponding forward and backward evolutions cancel exactly at zero detuning; relaxation, dephasing, waveform imbalance, and finite bandwidth make the cancellation incomplete in experiment. This echo-like structure suppresses resonant Rabi oscillations while preserving the off-resonant spectral response, producing a sharp depletion feature centered at the qubit transition frequency. The waveform Fourier spectrum in Fig. 1 has a null at zero baseband frequency, consistent with its vanishing signed area, but this linear waveform property is not itself the nonlinear qubit response. The resonance-centered depletion instead follows from the driven dynamics and is observed in the experimental spectroscopy maps.

This mechanism can be viewed as a Loschmidt-echo-like evolution, in which ideal reversibility occurs only on resonance, while detuning, decoherence, and experimental imperfections break the cancellation and restore excitation [17, 18]. Combined with the adiabatic following of the root-Lorentzian envelope [14, 15], this interference effect suppresses both power broadening and coherent Rabi oscillations, enabling narrow spectroscopy over a broad range of drive strengths with driven-spectroscopy pulses shorter than T_1. This sub-T_1 operating regime remains useful as qubit lifetimes increase because square and ordinary root-Lorentzian pulses can still exhibit coherent oscillatory distortions. The numerical model used for the experiment–theory comparison in Fig. 2 is detailed in the Supplemental Material [16].

Fig. 2 compares measured and simulated spectroscopy for ordinary root-Lorentzian and echo-root-Lorentzian pulses on the primary qubit. Both protocols use $L = 20$ μs, $c = 0.005$, and a calibrated peak-Rabi range extending to approximately 42.5 MHz. The left and right blocks span broad 50 MHz and narrow 1 MHz detuning domains, respectively. The upper row shows experiment and the lower row shows three-level simulations with matched pulse duration and cutoff. The simulated $|e\rangle$ populations are shown without fitted offsets or gains; the measurement uses two-state discrimination and does not independently resolve $|f\rangle$ population. For the tested 4–16 ns midpoint-smoothed echo pulses, simulations predict final leakage into $|f\rangle$ below 1% over the sampled near-resonance range (Supplemental Material [16]). The ordinary root-Lorentzian measurements retain strong amplitude-dependent coherent structure. The echo-root-Lorentzian data instead show a narrow resonance-centered depletion channel over the

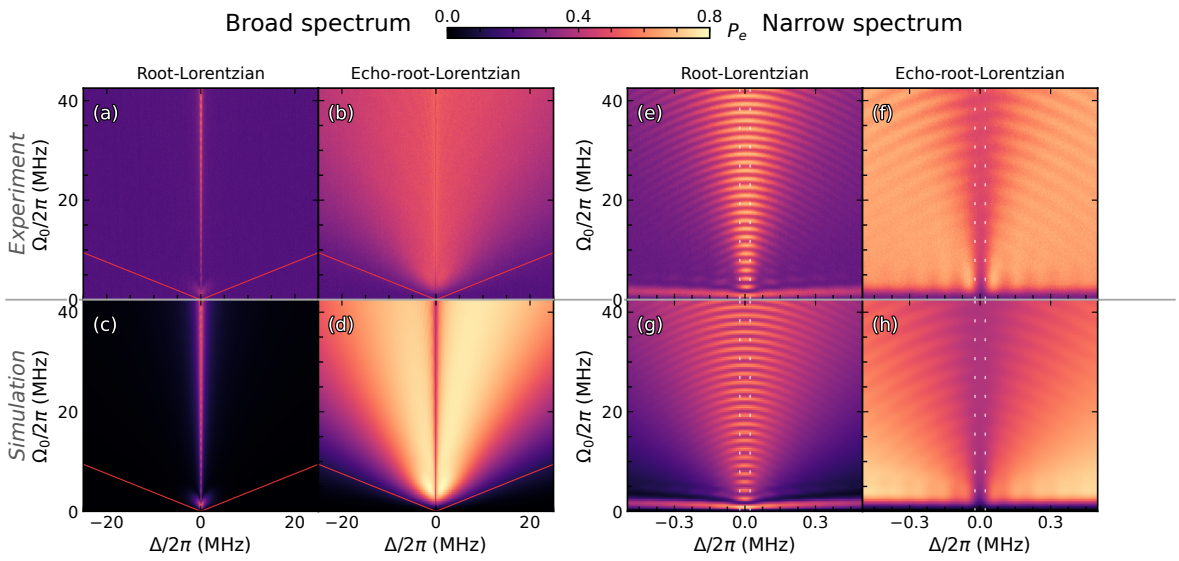


Figure 2. Experimental and simulated spectroscopy for $L = 20$ μs and $c = 0.005$. The left block (a)–(d) spans 50 MHz; the right block (e)–(h) resolves the central 1 MHz. Within each block, root-Lorentzian and echo-root-Lorentzian pulses occupy the left and right columns, respectively, with experiment above three-level Lindblad simulations. Red lines in (a)–(d) mark the linear high-drive half-width of the constant-drive reference, $\Delta/(2\pi) = \pm[\Omega_0/(2\pi)]\sqrt{T_1/T_2}$, using $T_2 = 7.31$ μs. Thin white dashed lines in (e)–(h) mark $\Delta/(2\pi) = \pm 1/(2\pi T_2)$, enclosing the coherence-limited FWHM $1/(\pi T_2) = 43.5$ kHz. Both pulse shapes produce much narrower central spectral features than the constant-drive reference. The root-Lorentzian response exhibits pronounced coherent oscillations, whereas the echo response retains a narrow resonance-centered depletion across the drive range. Residual high-drive oscillations in the echo response are consistent with AC-Stark phase accumulation, which is not reversed by the echo phase inversion.

measured drive range.

The Supplemental Material [16] provides a complementary fixed-amplitude comparison for the ordinary and echo protocols.

To identify the useful operating regime, we examine the linewidth and contrast of the echo response as functions of pulse duration and drive amplitude. We choose pulse durations shorter than T_1, but long enough that Fourier broadening does not dominate. The duration-dependent linewidth, contrast, and contrast-weighted resolution are reported in the Supplemental Material [16].

A second design constraint is the cutoff of the formally infinite-support pulse envelope. Truncation introduces a residual endpoint drive $\Omega_c = c\Omega_0$ and a corresponding cutoff-broadening contribution [15, 19, 20]. The relevant weak-saturation condition is $s_c = \Omega_c^2 T_1 T_2 \ll 1$, or $\Omega_c \ll 1/\sqrt{T_1 T_2}$; the inequality therefore points toward *smaller*, not larger, endpoint amplitudes. The relative cutoff c cannot be specified independently of Ω_0, T_1, and T_2, giving a three-dimensional design space of pulse duration, drive amplitude, and cutoff level.

The waveform itself offers a further control of the linewidth–contrast tradeoff. A derivative removal by adiabatic gate (DRAG) pulse adds a quadrature component proportional to the time derivative of the in-phase waveform to suppress leakage and drive-induced phase errors [6, 7]. Figure 3 compares three-level simulations for a 30 μs echo-root-Lorentzian pulse with $c = 0.001$: an uncorrected outer envelope, the same envelope with DRAG, and DRAG with a polynomial edge correction. All three use the same sine-smoothed phase jump $s(t)$ of duration $T_s = 16$ ns at the midpoint. The corrected pulse is

$$L_{\mathrm{floor}}^{(1/2)}(t) = \frac{L^{(1/2)}(t) - c}{1 - c}, \tag{4}$$

$$W_p(t) = \left[1 - \left(\frac{2t}{L}\right)^2\right]^p, \qquad p = 5, \tag{5}$$

$$s(t) = \begin{cases} +1, & t \le -T_s/2, \\ -\sin(\pi t/T_s), & |t| < T_s/2, \\ -1, & t \ge T_s/2, \end{cases} \tag{6}$$

$$L^{\mathrm{smooth}}(t) = L_{\mathrm{floor}}^{(1/2)}(t)\, s(t)\, W_p(t). \tag{7}$$

Here $|t| < L/2$, $\tau = (L/2)/\sqrt{c^{-2} - 1}$, and $W_p = 0$ outside the pulse. The floor correction removes the endpoint offset, and $W_p(t)$ smooths the edges. Thus, (a)–(b) isolate DRAG, while (b)–(c) isolate this edge correction. With $I(t) = \Omega_0 L^{\mathrm{smooth}}(t)$, the added DRAG quadrature is

$$Q(t) = -\frac{\beta}{|\alpha|}\frac{dI(t)}{dt}, \tag{8}$$

where $\alpha = \omega_{12} - \omega_{01}$ is the qubit anharmonicity. We use DRAG with $\beta = -0.5$ for near-optimal compensation of

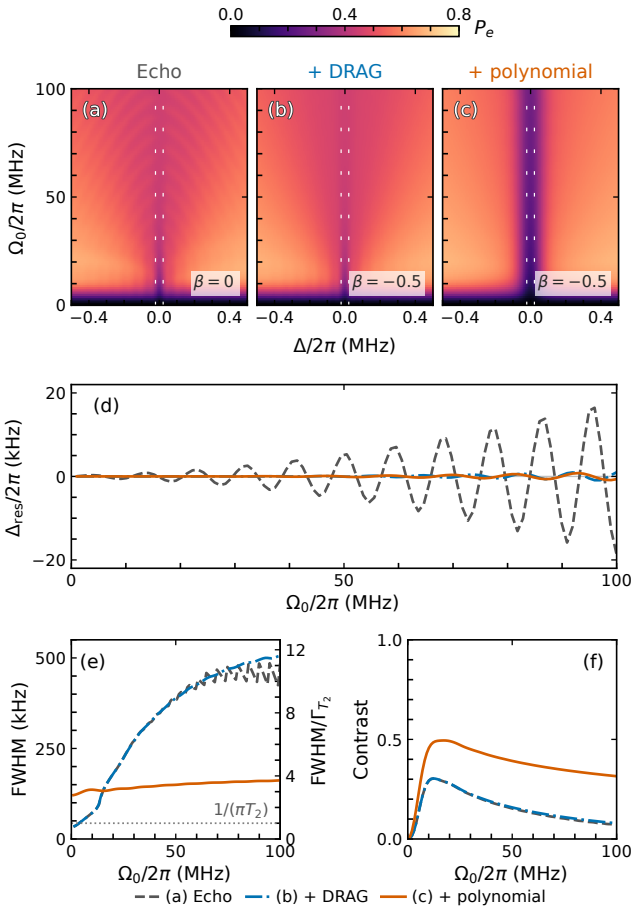


Figure 3. Simulated improvement through pulse shaping. (a)–(c) Final P_e for an echo-root-Lorentzian pulse (echo), echo with DRAG, and echo with DRAG plus cutoff subtraction and a polynomial window, respectively. All pulses have duration $L = 30$ μs and cutoff $c = 0.001$. White dashed lines bracket the coherence-limited linewidth. (d) Resonance-center detuning. (e) Dip FWHM. (f) Contrast C, the dip depth in excited-state probability. DRAG compensates drive-induced detuning, suppressing resonance motion as the amplitude varies. The polynomial correction smooths the pulse edges, reducing high-drive broadening and preserving greater dip contrast. Together, these corrections keep the resonance more stable and distinguishable over a broad drive-amplitude range.

AC-Stark-induced detuning [6, 7], as supported by the coefficient scan in the Supplemental Material [16].

DRAG suppresses amplitude-dependent resonance motion: the interpolated center RMS falls from 6.54 kHz for the echo to 0.31 kHz with DRAG and 0.29 kHz with DRAG plus the polynomial correction [Fig. 3(d)]. These sub-grid estimates quantify the saved maps; they do not independently establish sub-kHz numerical accuracy. DRAG does not, by itself, prevent the high-drive linewidth increase in this comparison. Adding the polynomial correction changes the FWHM at $\Omega_0/(2\pi) =$ 100 MHz from 506 to 162 kHz, while the probability contrast increases from 0.079 to 0.315. The corrected pulse maintains contrast above 0.31 across the sampled 10–100 MHz range. This improvement comes at the cost of a broader low-drive line, and the high-drive width remains above $1/(\pi T_2)$. Here contrast is the dip depth relative to the local background, expressed as a probability difference rather than normalized by measurement noise. These simulations establish a pulse-shaping tradeoff rather than an experimental sensitivity gain or a globally optimized waveform.

Figure 4 maps the joint dependence on cutoff and drive amplitude for the 30 μs polynomial-corrected DRAG echo. The white contours mark $C = 0.2$: the band between them has a dip depth exceeding 0.2 in excitation probability. Across this broad region of parameter space, the simulated response combines appreciable contrast with local FWHM values of 112–182 kHz, approximately 2.6–4.2 times the coherence reference Γ_{T_2}. The contours select contrast; the linewidth within that region is read independently from panel (a).

The useful band shifts toward smaller cutoff as drive increases, yet a single cutoff can retain both narrow linewidth and high contrast over a wide drive range. For example, at the sampled $c = 6.58 \times 10^{-4}$, the local FWHM remains below 155 kHz and contrast above 0.32 throughout the sampled 10–100 MHz interval. Here c sets the original root-Lorentzian core width even though the corrected envelope vanishes at its endpoints: changing c also changes τ. The full duration survey and numerical checks are given in the Supplemental Material [16].

In conclusion, we have demonstrated a narrow spectroscopic depletion feature that persists over more than two decades of peak drive. Measurements establish its linewidth, contrast, and drive-induced center motion, while numerical calculations show how DRAG and polynomial edge correction can improve high-drive linewidth and probability contrast. The experiment–simulation comparison in Fig. 2 supports the echo interpretation, while the duration–cutoff scan identifies the competing linewidth and contrast requirements. Phase-inverted shaped pulses thus provide a route to high-resolution qubit spectroscopy beyond weak drive.

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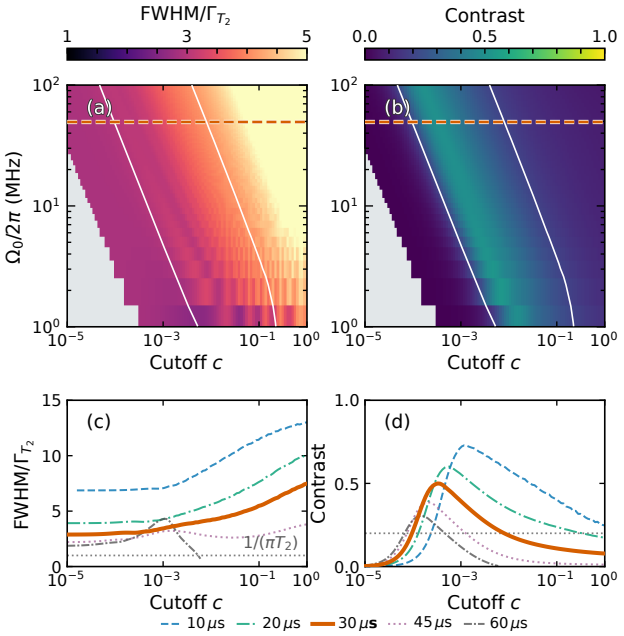


Figure 4. Simulated cutoff and duration dependence of the polynomial-corrected DRAG echo. (a),(b) Normalized dip FWHM and contrast C versus cutoff and drive amplitude for $L = 30$ μs. The white contours enclose the $C > 0.2$ band, where panel (a) also shows narrow dips. Cutoff controls the pulse-core width; this band balances spectral narrowing against loss of contrast, retaining a distinguishable resonance over a broad amplitude range. (c),(d) Cutoff slices at $\Omega_0/(2\pi) = 49.5$ MHz (orange dashed lines) for durations $L = 10$–60 μs. The thick orange 30 μs curves match the maps; dotted lines mark FWHM/$\Gamma_{T_2} = 1$ in (c) and $C = 0.2$ in (d).

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